

COMPUTER NETWORKING

Comprehensive Exam Revision Guide

Complete Revision Notes for All Chapters

CHAPTER 1: INTRODUCTION TO COMPUTER NETWORKS

1. Computer Network

★ **Exam Definition:**

A Computer Network is a collection of two or more computers/devices connected together to share data, resources, and services.

Key Points:

- Connects multiple devices.
- Shares data and files.
- Shares hardware (Printer, Scanner).
- Shares Internet connection.
- Enables communication.

 **Memory Trick: Network = Connect + Communicate + Share**

2. Advantages of Computer Network

Key Points:

- Resource Sharing
- File Sharing
- Fast Communication
- Cost Saving
- Centralized Management
- Easy Backup

 **Memory Trick: "Share More, Spend Less." (Network mainly exists for sharing.)**

3. Types of Networks

Definition: A small network around one person. Example: Bluetooth. Range: 1–10 m. Trick: P = Personal

Definition: Network inside one building. Example: School Lab. Range: Up to a few km. Trick: L = Local

Definition: Connects LANs within a city. Example: City Cable. Range: Entire City. Trick: M = Metropolitan

Definition: Over large geographical areas. Example: Internet. Trick: W = World

4. Network Topologies

Bus Topology: One main cable. Cheap. Backbone failure = Network fails. Trick: Bus = One Road

Star Topology: Connect to central Hub/Switch. Easy maintenance. Trick: Star = Center Device

Ring Topology: Forms a circle. Data moves ring-wise. Trick: Ring = Circle

Mesh Topology: Every device connected to every other. Reliable but expensive. Trick: Mesh = Everyone Connected

Tree Topology: Star + Bus combo. Hierarchical. Trick: Tree = Branches

5. OSI Model

★ **Exam Definition:**
Seven-layer reference model explaining data travel.

 **Memory Trick: All People Seem To Need Data Processing (Application, Presentation, Session, Transport, Network, Data Link, Physical)**

Layer	Data Unit	Device
App/Pres/Sess	Data	-
Transport	Segment	-
Network	Packet	Router
Data Link	Frame	Switch/Bridge
Physical	Bits	Hub/Repeater

6. TCP/IP Model

Layers: Application, Transport, Internet, Network Access. OSI is Theory, TCP/IP is Practical.

7. Encapsulation & Decapsulation

Encapsulation: Headers added moving down. Decapsulation: Headers removed moving up.



Memory Trick: Encapsulation = Packing; Decapsulation = Unpacking

CHAPTER 2: PHYSICAL LAYER (LAST-MINUTE REVISION)

Physical Layer (Definition)

★ **Exam Definition:**

The Physical Layer is the 1st layer of the OSI Model, responsible for transmitting raw bits through a physical medium.

Key Points:

- 1st layer of OSI
- Data Unit = Bits
- Defines cables/signals
- No error checking

 **Memory Trick: Physical = Wire + Signal + Bits**

1. Transmission Media

Twisted Pair: Two insulated copper wires. Used for Telephone/LAN. Cheap. Trick: Telephone Cable

Coaxial Cable: Central conductor + shielding. Used for TV/Broadband. Trick: TV Cable

Optical Fiber: Uses light. High speed, long distance. Trick: Light = Fastest

Radio Waves: Long distance, broadcasting. Trick: Radio = Broadcast

Microwaves: Satellite/Mobile communication. Trick: Microwave = Satellite

Infrared: Short range. Trick: TV Remote

2. Encoding

Converting binary to signals.

NRZ: No Return to Zero. 1 = High, 0 = Low. Synchronization issues.

Manchester: Middle change in every bit. Reliable synchronization. Trick: Manchester = Middle Change

3. Multiplexing

FDM: Frequency Division. Different Frequencies (Radio). Trick: F = Frequency

TDM: Time Division. Different Time Slots (Ticket Counter). Trick: T = Time

WDM: Wavelength Division. Uses Light (Fiber). Trick: W = Wavelength

4. Switching Techniques

Circuit Switching: Dedicated path (Telephone).

Packet Switching: Divided into packets (Internet).

Message Switching: Store and Forward.

CHAPTER 3: DATA LINK LAYER (LAST-MINUTE REVISION)

Data Link Layer (Definition)

★ **Exam Definition:**

The 2nd layer of OSI, responsible for node-to-node communication, framing, and error detection.

📌 **Key Points:**

- Data Unit = Frame
- Uses MAC Address
- Framing, Flow Control, Error Detection

🧠 **Memory Trick: Data Link = Frame + Flow + Error + MAC**

1. Flow Control

Stop-and-Wait: One frame, wait for ACK. Simple but slow.

Sliding Window: Multiple frames without ACK. Faster efficiency.

2. Error Detection

Parity Check: Count 1s. Simple but limited.

CRC: Cyclic Redundancy Check. Polynomial division. Accurate.

Checksum: Add + Complement.

3. Error Control (ARQ)

Stop-and-Wait ARQ: Send -> Wait -> ACK.

Go-Back-N: Resend from error onwards.

Selective Repeat: Only lost frame returns.

4. Protocols & Standards

HDLC: Reliable bit-oriented. ALOHA: Pure (anytime) vs Slotted (fixed slots).

Key Points:

- IEEE 802.3 = Ethernet
- IEEE 802.4 = Token Bus
- IEEE 802.5 = Token Ring

CHAPTER 4: NETWORK LAYER (LAST-MINUTE REVISION)

Network Layer (Definition)

★ **Exam Definition:**

The 3rd layer of OSI, responsible for IP addressing, routing, and packet forwarding.

📌 **Key Points:**

- Data Unit = Packet
- Uses IP Address
- Uses Routers

1. Routing Algorithms

Shortest Path: Minimum distance.

Distance Vector: Hop counts (Used by RIP).

Broadcast: One to all.

Multicast: One to selected group.

2. IP Addressing

IPv4 (32-bit): Classes A(1-126), B(128-191), C(192-223), D(Multicast), E(Research).

IPv6 (128-bit): Future, colossal space, hexadecimal.

3. Protocols

RIP: Hop counts (max 15). Small networks.

OSPF: Cost based. Large networks.

BGP: Internet routing.

NAT: Private <-> Public conversion.

ICMP: Ping and error messages.

CHAPTER 5: TRANSPORT LAYER (LAST-MINUTE REVISION)

Transport Layer (Definition)

★ **Exam Definition:**

The 4th layer of OSI, providing end-to-end communication between apps.

📌 **Key Points:**

- Data Unit = Segment
- Segmentation & Reassembly
- Port Addressing (0-65535)
- Flow & Error Control

1. TCP vs UDP

TCP: Connection-oriented, Reliable, 3-Way Handshake (SYN -> SYN+ACK -> ACK).

UDP: Connectionless, Fast, No ACK. Used for Video/Gaming.

2. Congestion Control

Leaky Bucket: Fixed output rate. No burst. Drops packets if full.

Token Bucket: Tokens for permission. Allows burst data.

3. Common Ports

📌 **Key Points:**

- HTTP: 80
- HTTPS: 443

- FTP: 21
- SMTP: 25
- DNS: 53

CHAPTER 6: APPLICATION LAYER (LAST-MINUTE REVISION)

Application Layer (Definition)

★ *Exam Definition:*

The 7th layer of OSI, closest to the user, providing services directly to software.

1. Networking Protocols

DNS: Domain Name -> IP. (Phonebook of Internet).

HTTP/HTTPS: Web protocol (Unsecure vs Encrypted).

FTP: File Transfer.

SMTP: Send Email.

POP3: Pull/Download Mail (Deletes from server).

IMAP: Mail remains on server (Sync across devices).

2. Security & Management

SSH vs Telnet: Both Remote Login. SSH is secure/encrypted, Telnet is clear text.

SNMP: Network Manager (Monitoring/Detecting problems).

NFS vs SMB: NFS is Linux file sharing. SMB is Windows (File+Printer).

CHAPTER 7: EMERGING NETWORKING CONCEPTS

Modern Networking

★ *Exam Definition:*

Technologies making networks faster, smarter, and centralized.

1. Cloud & IoT

Cloud Networking: Storage and services on remote servers. (Access Anywhere).

IoT (Internet of Things): Network of smart physical devices (Bulbs, Watches, Cars).

2. IoT Protocols

MQTT: Broker-based (WhatsApp Group Admin logic). Pub-Sub model.

CoAP: Direct Client-Server (Phone Call logic). Lightweight.

3. SDN (Software Defined Networking)

Separates Control Plane (Decision Maker) from Data Plane (Forwards packets). Centralized control via SDN Controller.



Memory Trick: SDN Controller = Brain; Routers = Workers.